

Defays2026_Polimi

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General-purpose geothermal surface plant simulation model using TESPpy (ORCs). Developed for a Master's thesis at the University of Liège (ULiège), during an internship at Politecnico di Milano (Polimi).

Overview

This repository contains a TESPpy-based simulation and optimization framework for geothermal Organic Rankine Cycle (ORC) power plants, developed incrementally across three thesis chapters:

- **Chapter 1** — building-block validation models (heat pump, fan, heat exchanger comparison, ORC subsystem interface).
- **Chapter 2** — theoretical/literature background; no code associated with this chapter, so it is not represented in this repository.
- **Chapter 3** — low-enthalpy ORC case studies.
- **Chapter 4** — high-enthalpy two-phase geothermal ORC: the core thesis model, including working-fluid selection, single-point optimization, parametric sweeps across resource temperatures, and exergy analysis.

Repository structure

```
Chapter1_Cases/
├── Fan/                # Fan model
├── HP/                 # Heat pump model, pymoo
└── optimization, exergy analysis
    ├── HXs_comparison/ # Heat exchanger model comparison
    └── Subsystem_Interface/ # ORC preheater/recuperator subsystem
        tests

Chapter3_Cases/
├── Case1/              # Low-enthalpy ORC, inputs from
└── documentation
    ├── Case2/          # PropsSI vs. TESPpy comparison
    └── Case3/           # Low-binary ORC test

Chapter4_Cases/
├── Chapter4_Final_version/ # Final high-enthalpy geothermal
└── ORC model
    ├── v3_HighEnthalpyGeoORC_Template.py #
    └── Simulation template
        ├── v3_HighEnthalpyGeoORC_Optimize.py # Single-
        └── point optimization (pymoo)
            ├── v3_HighEnthalpyGeoORC_Parametric_Optimize_Parallel.py #
            └── Parametric sweep over T_geo
                ├── v3_HighEnthalpyGeoORC_Exergy.py # Exergy
                └── analysis (ExerPy)
                    ├── v3_HighEnthalpyGeoORC_Diagrams.py # Single-
                    └── case diagrams
                        ├── v3_HighEnthalpyGeoORC_parametric_Diagrams.py #
                        └── Parametric sweep diagrams
```

```

|   ├── CSVs/                                     #
|   └── Optimization logs and results
|       ├── Diagrams/                             # Generated
|       └── SVG figures
|           ├── Chapter4_drum_version/             # Earlier drum-based architecture
|           │   (kept for reference)
|           └── chapter4_fluids/                   # Working-fluid selection study
|
| functions/
|   ├── Fluprodia_diagrams_f.py                   # Fluid property (T-s, log p-h)
|   └── diagram helpers
|       ├── HX_diagrams_f.py                     # Heat exchanger Q-T diagram
|       └── helpers

```

Requirements

Python 3.12. **TESPy's API changed across the thesis's development timeline (v0.9.12 → v0.10.2), so each chapter needs its own TESP version** — there isn't a single `requirements.txt` that works for the whole repository. Install the one matching the chapter you want to run:

Chapter	TESPy version	Requirements file
Chapter 1	0.9.16 <i>(reconstructed from memory — verify before relying on it)</i>	Chapter1_Cases_requirements.txt
Chapter 3	0.9.14	Chapter3_Cases_requirements.txt
Chapter 4 (main model)	0.10.2	requirements.txt

```

# Chapter 4 (main model)
pip install -r requirements.txt

# Chapter 1
pip install -r Chapter1_Cases_requirements.txt

# Chapter 3
pip install -r Chapter3_Cases_requirements.txt

```

Since the TESP versions are incompatible with each other, use a separate virtual environment per chapter if you need to run more than one.

Key dependencies: [TESPy](#) (thermal system simulation), [ExerPy](#) (exergy analysis), [CoolProp](#) (fluid properties), [pymoo](#) (multi-objective/differential-evolution optimization), [fluprodia](#) (fluid property diagrams, Chapter 1 only).

Usage — Chapter 4 (main model)

The scripts in `Chapter4_Cases/Chapter4_Final_version/` are meant to be run in this order:

1. `v3_HighEnthalpyGeoORC_Template.py` — defines and solves the base plant model for a single operating point.
2. `v3_HighEnthalpyGeoORC_Optimize.py` — single-point differential-evolution optimization (`p_evap` , `dT_cd`) for a given resource temperature.

3. `v3_HighEnthalpyGeoORC_Parametric_Optimize_Parallel.py` — repeats the optimization across a sweep of geothermal resource temperatures.
4. `v3_HighEnthalpyGeoORC_Exergy.py` — exergy analysis and component-level exergy-destruction comparison across working fluids.
5. `v3_HighEnthalpyGeoORC_Diagrams.py` / `v3_HighEnthalpyGeoORC_parametric_Diagrams.py` — generate the SVG figures used in the thesis.

Author

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